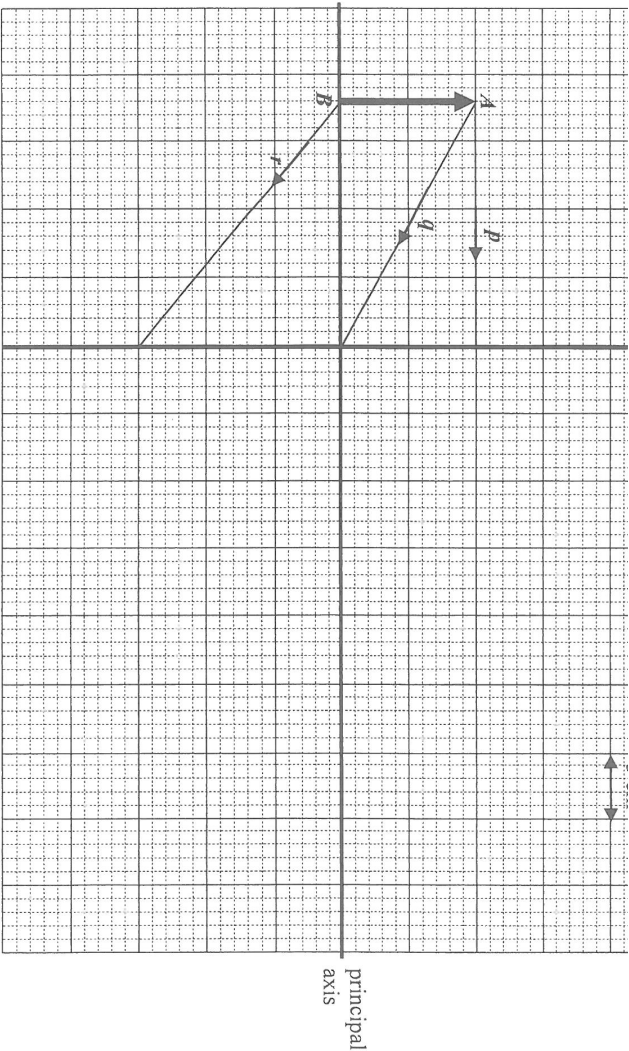


Answers written in the margins will not be marked.



- (ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of p , q and r . (3 marks)
- (iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

Focal length =

- (iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

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Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

- (ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed. (2 marks)

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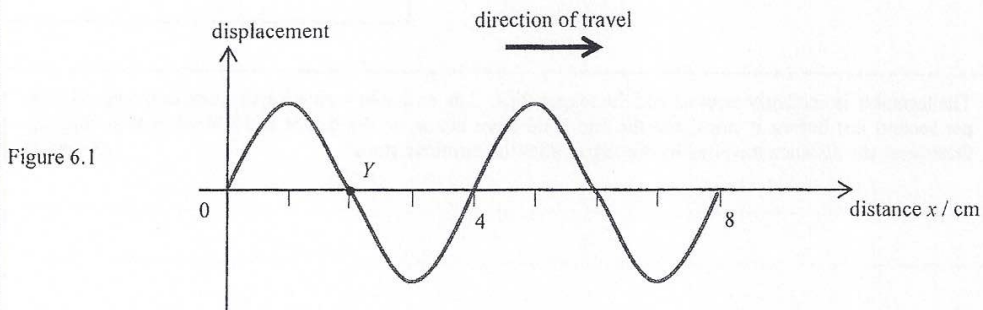
- (c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

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Answers written in the margins will not be marked.

6. (a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.



- (i) Determine the wave speed of the water wave. (2 marks)

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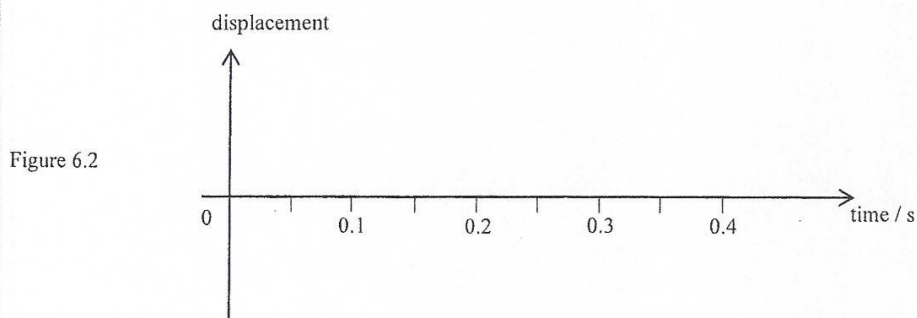
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- (ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

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- (iii) Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)

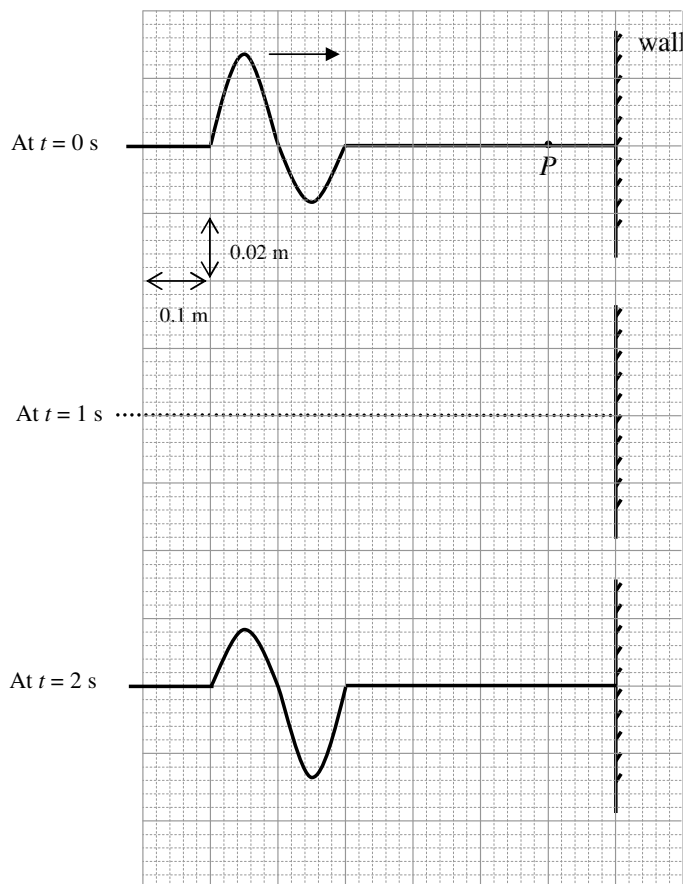


Answers written in the margins will not be marked.

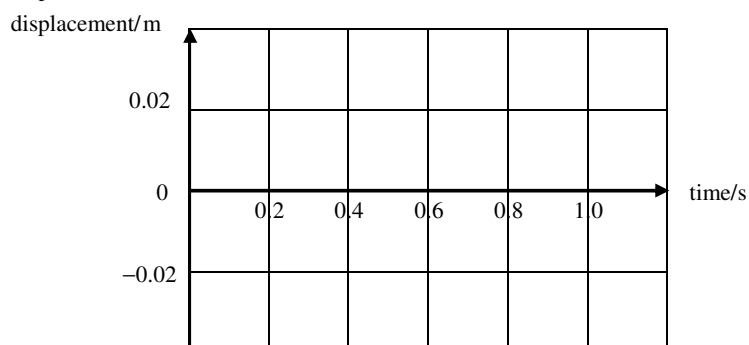
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4. One end of a piece of string is fixed to a wall. A wave pulse travels along the string at a speed of 0.5 m s^{-1} towards the fixed end. Figure 4.1 shows the string at time $t = 0 \text{ s}$ and $t = 2 \text{ s}$.

Figure 4.1



- (a) On Figure 4.1, draw the shape of the wave pulse at $t = 1 \text{ s}$. (1 mark)
- (b) Sketch a graph of the displacement of point P on the string at a distance of 0.1 m from the wall during the period $t = 0 \text{ s}$ to $t = 1 \text{ s}$. (2 marks)



Answers written in the margins will not be marked.